

Full Name: _____
EEL 3135 – Classwork #06

Milestone:

- Complete any 2 questions to receive credit.

Question #1: Given:

$$x(t) = 8e^{j(6t-\pi/3)} + 8e^{-j(6t-\pi/3)}$$

- (a) Use Euler's identity to convert $x(t)$ into its real sinusoidal form.
- (b) Identify the amplitude, phase shift, and frequency of the resulting cosine wave.
- (c) Generate and label a MATLAB plot showing two full periods of $x(t)$.

Question #2: You are given the following frequency-domain information for $y(t)$:

$$\omega_0 = 3 \text{ rad/s}, \quad a_0 = 2, \quad a_{\pm 1} = 1.5e^{\mp j\pi/2}, \quad a_{\pm 2} = 0.5e^{\mp j\pi/4}$$

(a) Write $y(t)$ as a sum of complex exponentials.

(b) Rewrite $y(t)$ using cosine terms (real Fourier series).

(c) Plot $y(t)$ over two fundamental periods in MATLAB with proper axis labels.

Question #3: Consider the signal

$$x(t) = 10 \cos(400\pi t) + 4 \cos(700\pi t)$$

- (a) Determine the highest frequency component present in $x(t)$.

- (b) Find the minimum sampling frequency (Nyquist rate) required to avoid aliasing.

- (c) If the sampling rate is $f_s = 500$ Hz, compute the aliased frequency (if any) for each cosine term and determine the resulting discrete-time frequencies.

Question #4: Consider the continuous-time signal

$$x(t) = 6 \cos(150\pi t) - 4 \cos(250\pi t)$$

- (a) Determine the highest frequency component in $x(t)$.
- (b) Compute the Nyquist rate.
- (c) If the signal is sampled at $f_s = 180$ Hz, state whether aliasing will occur and explain why.